



FIRST SEMESTER 2026-27
Course Handout Part II

01-08-2026

In addition to Part-I (General Handout for all courses appended to the time table) this portion gives further specific details regarding the course.

Course No. : MAC F313
Course Title : Statistical Data Analysis
Instructor-in-Charge : Dr. Harsha K V, Department of Mathematics
Instructors : Harsha K V
Tutorial Instructors : Harsha K V, Debismita Nayak

Scope and Objective of the Course: The course introduces fundamental methods for statistical data analysis, simulation, regression, classification, and Bayesian inference, with an emphasis on practical implementation using R for analyzing real-world datasets.

Textbook:

- T1. James, G., Witten, D., Hastie, T., and Tibshirani, R.,** An Introduction to Statistical Learning with Applications in R, Springer, New York, 2013.
T2. Ross, S. M., Simulation, Academic Press, Fourth Edition, 2006.
T3. Anderson, D. R., Sweeney, D. J., Williams, T. A., Camm, J. D., and Cochran, J. J., Statistics for Business and Economics, 13th Edition, Cengage Learning, 2014.

Reference books:

- R1. Rencher, A. C., and Schaalje, G. B.,** Linear Models in Statistics, 2nd Edition, John Wiley & Sons, 2008.
R2. Ross, S. M., Introductory Statistics, 3rd Edition, Academic Press, 2010.
R3. Robert, C. P., and Casella, G., Introducing Monte Carlo Methods with R, Springer, 2010.

Course Plan:

Lecture No.	Learning objectives	Topics to be covered	Chapter in the Text Book
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1-3	Summarize, describe, and visualize one- and two-dimensional data	Introduction to statistics, types of data; graphical representation of 1-D data (histograms, box plots, stem-leaf); graphical representation of 2-D data (scatter plots, Q-Q plots)	T3: Ch.1–3; R2: Ch 2-3
4-8	Simulate random variates from standard distributions using general-purpose techniques	Simulating Binomial, Poisson, Normal, Gamma, and Exponential random variables; inverse transform method; acceptance–rejection method	T2: Ch.4–5
9-13	Test distributional assumptions and relationships between variables	Chi-square goodness of fit test; Kolmogorov–Smirnov test; contingency tables and tests for independence	T2: Ch.5; R3: Ch.2-3
14–19	Build, estimate, and validate a simple linear regression model	Simple linear regression: model formulation, least squares estimation, MLE approach, inference on coefficients, residual analysis, model adequacy	T1: Ch.3, T3: Ch. 14 R1: Ch. 6
20-24	Extend regression to multiple predictors and select relevant variables	Multiple linear regression: matrix formulation, inference, multicollinearity; variable selection	T1: Ch.3 T3: Ch. 15 R1: Ch. 7,9
25-30	Model and classify data using generalized linear and discriminant methods	GLM framework and link functions; logistic regression (estimation & inference); LDA; QDA; comparison of classifiers	T1: Ch.4 R1: Ch.18
31-35	Use simulation for estimation and integration problems	Monte Carlo simulation principles; MC integration; variance reduction; importance sampling; applications	T2: Ch.7–8 R3: Ch. 4-6
36-41	Apply Bayesian reasoning and compare it with frequentist inference	Bayes' theorem; likelihood, posterior; conjugate priors, and Bayesian learning; frequentist vs. Bayesian comparison; simple implementation	T3: Ch.4 R3: Ch.1-2

Lab Component Details:

Experiment No.	Title	Lab hours
Exp. 1	Graphical Representation of One- and Two-Dimensional Data	Lab 1-2
Exp. 2	Simulation of Random Variates (Direct, Inverse Transform, Accept-Reject)	Lab 3-4
Exp. 3	Goodness of Fit Tests and Tests of Independence	Lab 5-6
Exp. 4	Simple and Multiple Linear Regression with Diagnostics	Lab 7-8
Exp. 5	Classification- Logistic Regression, LDA, QDA	Lab 9-10
Exp. 6	Monte Carlo Methods and Bayesian Estimation	Lab 10-12

Evaluation Scheme:

Component	Duration	Weightage (%)	Date & Time	Nature of Component
Mid-Sem	90 minutes	30		Closed Book
Lab Assignment in R		10	To be announced through LMS.	Open Book
Continuous Evaluation Tests		10	To be announced through LMS.	Open Book
Quizzes (2 Nos.)	30 minutes for each	5+5	To be announced through LMS.	Closed Book
Comprehensive	3 Hours	40		Closed Book



Note:

1. Total marks with all the evaluation components taken together will be 100.
2. Performance of continuous evaluation tests will be computed using the **best (N-1) out of N** tests, where **N>3**.

Chamber Consultation Hour: To be announced in class.

Notices: Students will be notified only through LMS.

Make-up Policy: Make-up for any component of evaluation will be given only in genuine cases of absence. [Prior permission through Email is required]

Academic Honesty and Integrity Policy: Academic honesty and integrity are to be maintained by all the students throughout the semester and no type of academic dishonesty is acceptable.

INSTRUCTOR-IN-CHARGE
MAC F313

